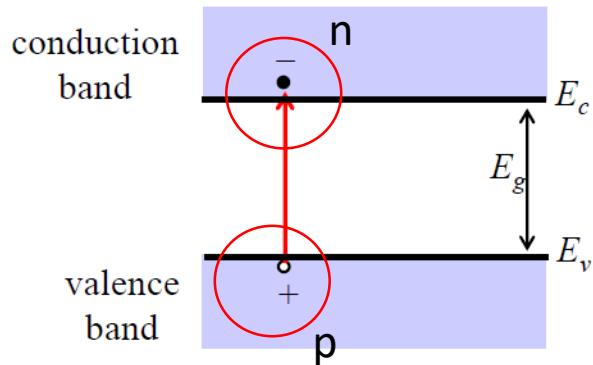


Intrinsic Semiconductor

No doping

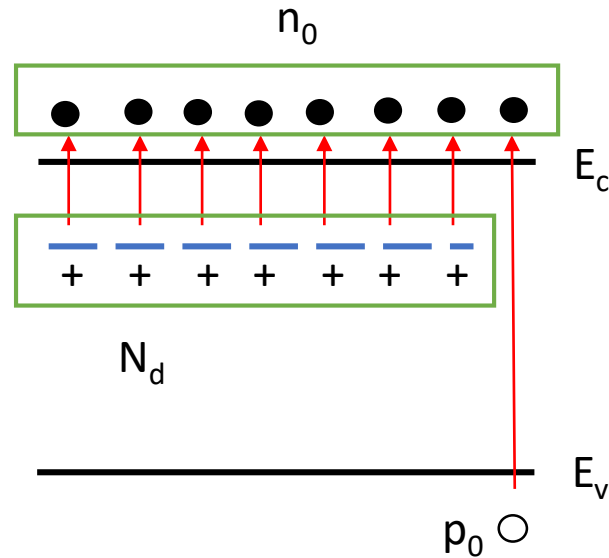


At thermal equilibrium

$$n_0 = p_0 = n_i$$

n-type

Doped with group V materials
(P, As, Sb)



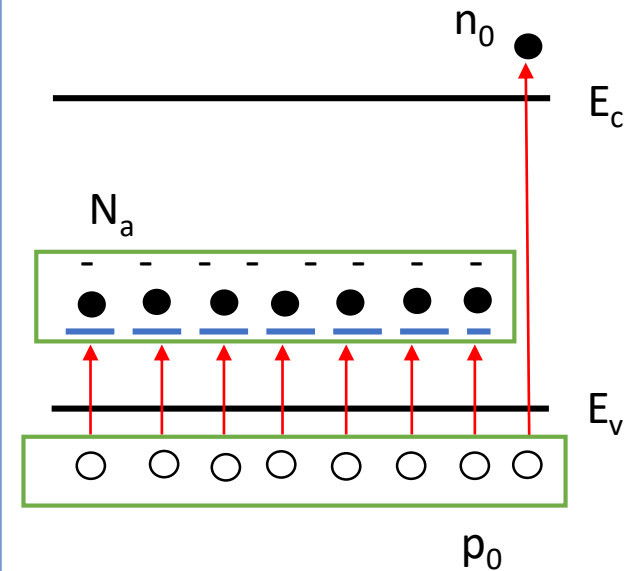
At $k_B T \gg$ ionization energy

$$n_0 = N_d + p_0$$

$$n_0 > p_0$$

p-type

Doped with group III materials
(B, Al, Ga)



At $k_B T \gg$ ionization energy

$$p_0 = N_a + n_0$$

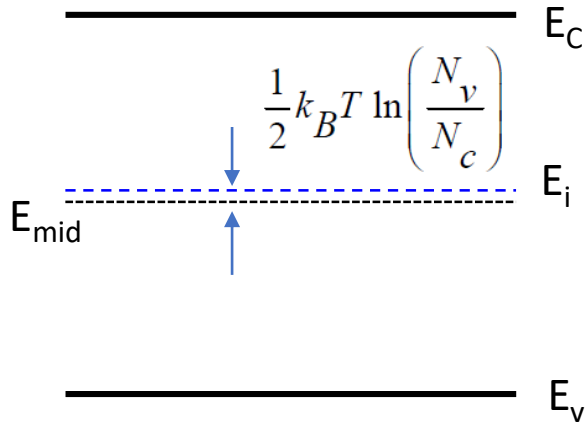
$$p_0 > n_0$$

Intrinsic Semiconductor

$$n_0 p_0 = n_i^2$$

$$n_i = N_c \exp\left[-\frac{E_c - E_i}{k_B T}\right]$$

$$p_i = N_v \exp\left[-\frac{E_i - E_v}{k_B T}\right]$$

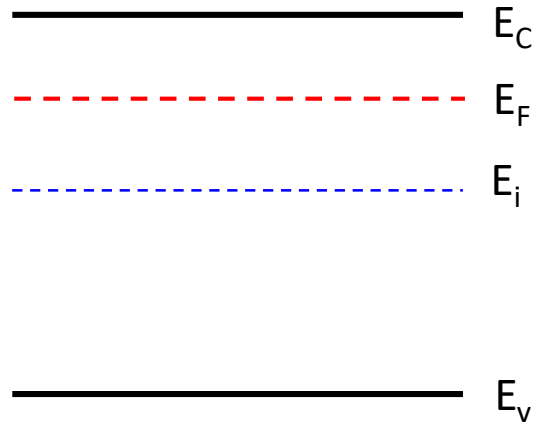


n-type

$$n_0 p_0 = n_i^2$$

$$n_0 = N_c \exp\left[-\frac{E_c - E_F}{k_B T}\right]$$

$$n_0 = n_i \exp\left[\frac{E_F - E_i}{k_B T}\right]$$

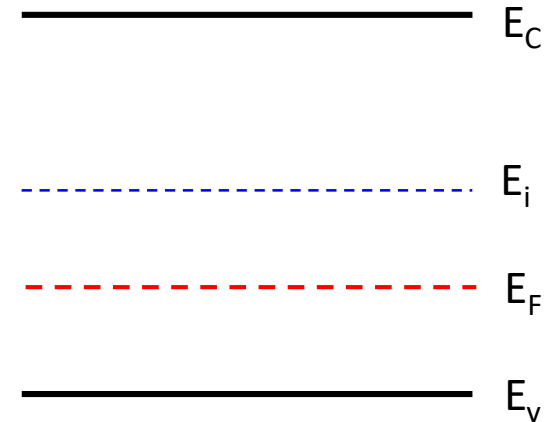


p-type

$$n_0 p_0 = n_i^2$$

$$p_0 = N_v \exp\left[-\frac{E_F - E_v}{k_B T}\right]$$

$$p_0 = n_i \exp\left[\frac{E_i - E_F}{k_B T}\right]$$



Intrinsic Semiconductor

$$n_i = \sqrt{N_c N_v} \exp\left[-\frac{E_g}{2k_B T}\right]$$

n-type

If $N_d \gg n_i$, $n_0 = N_d$

Otherwise

$$n_0 = \frac{N_d + \sqrt{N_d^2 + 4n_i^2}}{2}$$

p-type

If $N_a \gg n_i$, $p_0 = N_a$

Otherwise

$$p_0 = \frac{N_a + \sqrt{N_a^2 + 4n_i^2}}{2}$$

Compensated Semiconductor

$$n_0 = \frac{(N_d - N_a) + \sqrt{(N_d - N_a)^2 + 4n_i^2}}{2}$$

If $N_d - N_a \gg n_i$, $n_0 = N_d - N_a$

$$p_0 = \frac{(N_a - N_d) + \sqrt{(N_a - N_d)^2 + 4n_i^2}}{2}$$

If $N_a - N_d \gg n_i$, $p_0 = N_a - N_d$